

## 1. Description

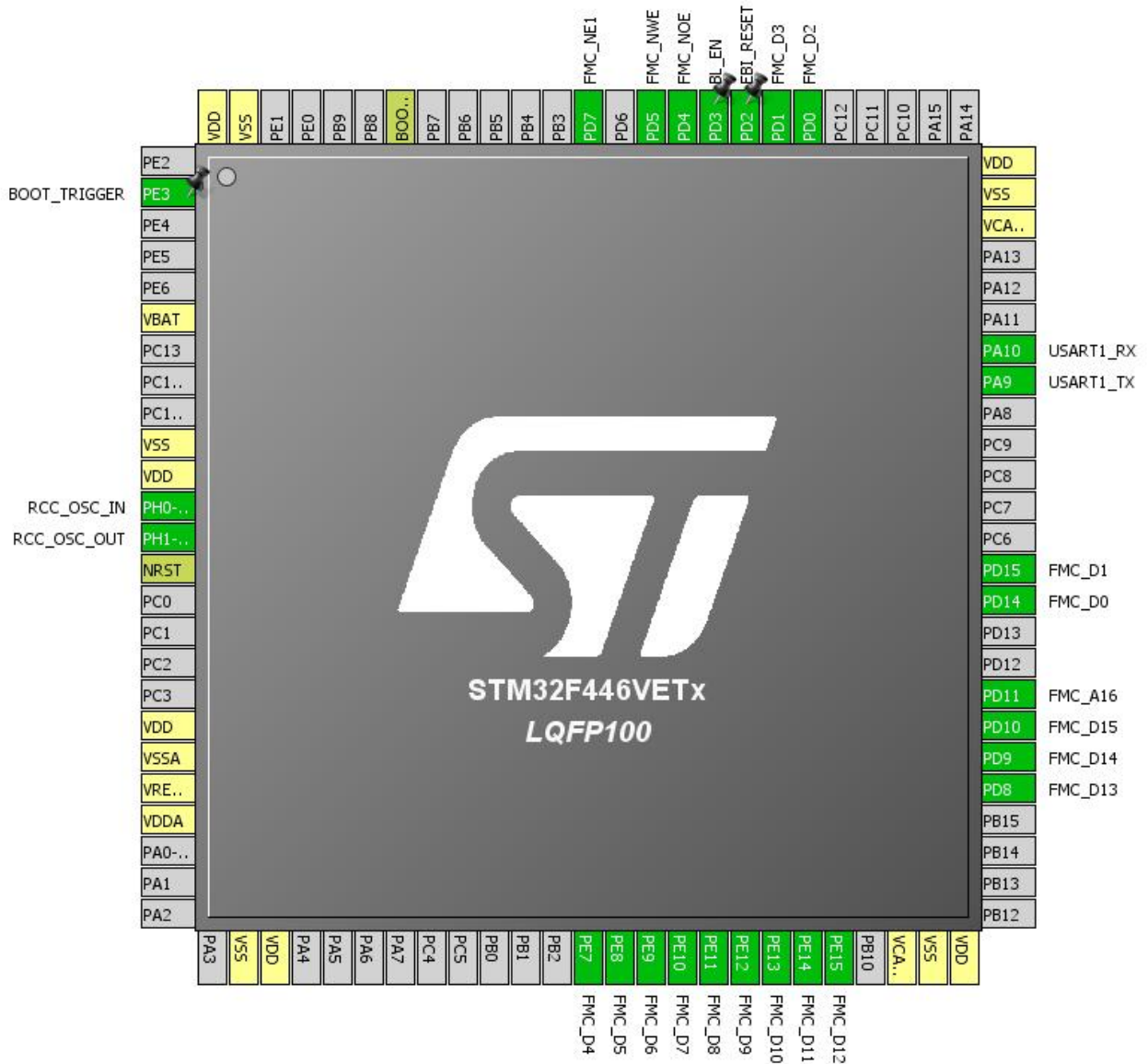
### 1.1. Project

|                 |                    |
|-----------------|--------------------|
| Project Name    | bootloader_st      |
| Board Name      | bootloader_st      |
| Generated with: | STM32CubeMX 4.21.0 |
| Date            | 05/22/2017         |

### 1.2. MCU

|                |               |
|----------------|---------------|
| MCU Series     | STM32F4       |
| MCU Line       | STM32F446     |
| MCU name       | STM32F446VETx |
| MCU Package    | LQFP100       |
| MCU Pin number | 100           |

## 2. Pinout Configuration



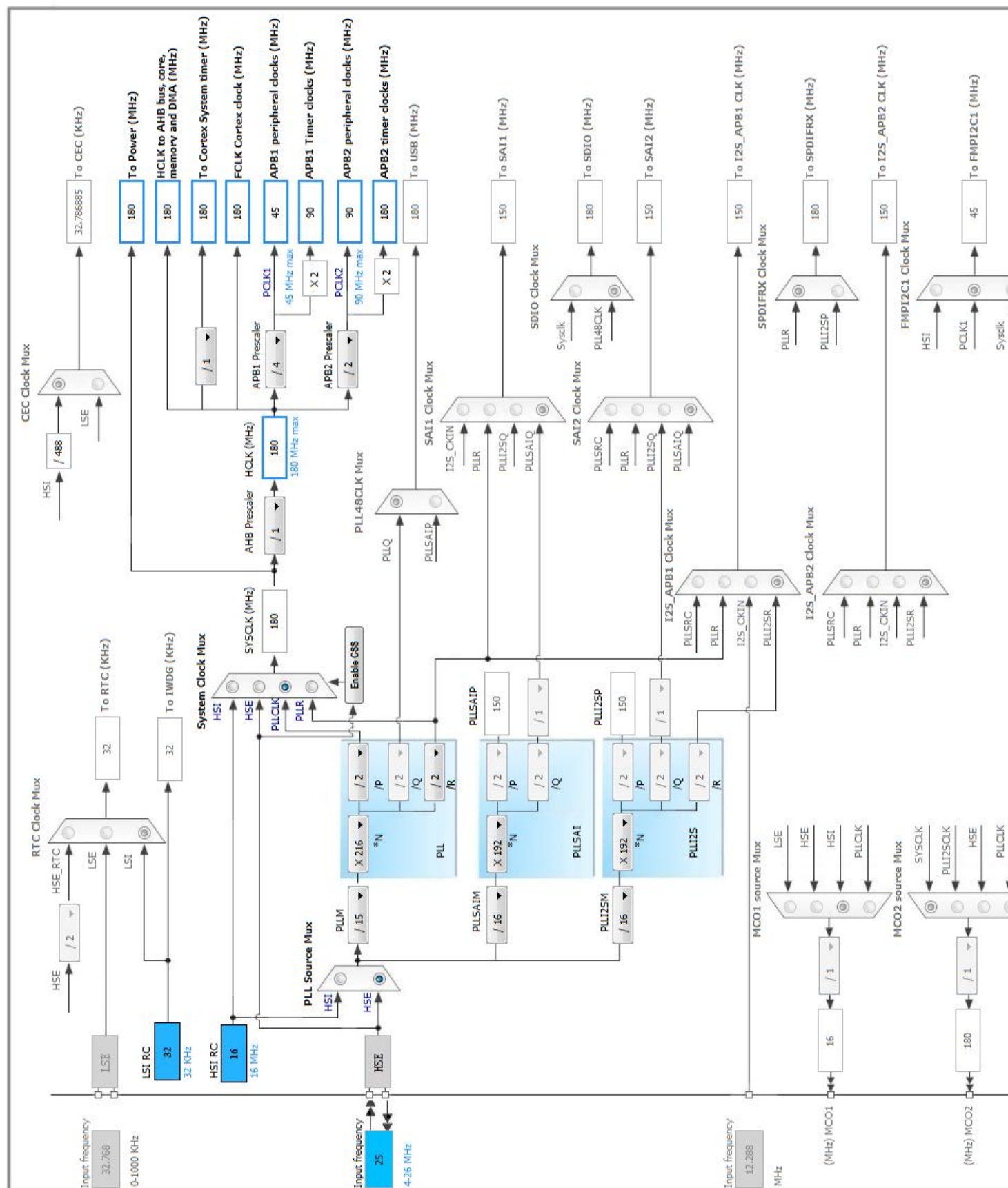
### 3. Pins Configuration

| Pin Number<br>LQFP100 | Pin Name<br>(function after<br>reset) | Pin Type | Alternate<br>Function(s) | Label        |
|-----------------------|---------------------------------------|----------|--------------------------|--------------|
| 2                     | PE3 *                                 | I/O      | GPIO_Input               | BOOT_TRIGGER |
| 6                     | VBAT                                  | Power    |                          |              |
| 10                    | VSS                                   | Power    |                          |              |
| 11                    | VDD                                   | Power    |                          |              |
| 12                    | PH0-OSC_IN                            | I/O      | RCC_OSC_IN               |              |
| 13                    | PH1-OSC_OUT                           | I/O      | RCC_OSC_OUT              |              |
| 14                    | NRST                                  | Reset    |                          |              |
| 19                    | VDD                                   | Power    |                          |              |
| 20                    | VSSA                                  | Power    |                          |              |
| 21                    | VREF+                                 | Power    |                          |              |
| 22                    | VDDA                                  | Power    |                          |              |
| 27                    | VSS                                   | Power    |                          |              |
| 28                    | VDD                                   | Power    |                          |              |
| 38                    | PE7                                   | I/O      | FMC_D4                   |              |
| 39                    | PE8                                   | I/O      | FMC_D5                   |              |
| 40                    | PE9                                   | I/O      | FMC_D6                   |              |
| 41                    | PE10                                  | I/O      | FMC_D7                   |              |
| 42                    | PE11                                  | I/O      | FMC_D8                   |              |
| 43                    | PE12                                  | I/O      | FMC_D9                   |              |
| 44                    | PE13                                  | I/O      | FMC_D10                  |              |
| 45                    | PE14                                  | I/O      | FMC_D11                  |              |
| 46                    | PE15                                  | I/O      | FMC_D12                  |              |
| 48                    | VCAP_1                                | Power    |                          |              |
| 49                    | VSS                                   | Power    |                          |              |
| 50                    | VDD                                   | Power    |                          |              |
| 55                    | PD8                                   | I/O      | FMC_D13                  |              |
| 56                    | PD9                                   | I/O      | FMC_D14                  |              |
| 57                    | PD10                                  | I/O      | FMC_D15                  |              |
| 58                    | PD11                                  | I/O      | FMC_A16                  |              |
| 61                    | PD14                                  | I/O      | FMC_D0                   |              |
| 62                    | PD15                                  | I/O      | FMC_D1                   |              |
| 68                    | PA9                                   | I/O      | USART1_TX                |              |
| 69                    | PA10                                  | I/O      | USART1_RX                |              |
| 73                    | VCAP_2                                | Power    |                          |              |
| 74                    | VSS                                   | Power    |                          |              |
| 75                    | VDD                                   | Power    |                          |              |

| Pin Number<br>LQFP100 | Pin Name<br>(function after<br>reset) | Pin Type | Alternate<br>Function(s) | Label     |
|-----------------------|---------------------------------------|----------|--------------------------|-----------|
| 81                    | PD0                                   | I/O      | FMC_D2                   |           |
| 82                    | PD1                                   | I/O      | FMC_D3                   |           |
| 83                    | PD2 *                                 | I/O      | GPIO_Output              | EBI_RESET |
| 84                    | PD3 *                                 | I/O      | GPIO_Output              | BL_EN     |
| 85                    | PD4                                   | I/O      | FMC_NOE                  |           |
| 86                    | PD5                                   | I/O      | FMC_NWE                  |           |
| 88                    | PD7                                   | I/O      | FMC_NE1                  |           |
| 94                    | BOOT0                                 | Boot     |                          |           |
| 99                    | VSS                                   | Power    |                          |           |
| 100                   | VDD                                   | Power    |                          |           |

\* The pin is affected with an I/O function

## 4. Clock Tree Configuration



## 5. IPs and Middleware Configuration

### 5.1. FMC

#### NOR Flash/PSRAM/SRAM/ROM/LCD 1

**Chip Select:** set

**Memory type:** LCD Interface

**LCD Register Select:** A16

**Data:** 16 bits

#### 5.1.1. NOR/PSRAM 1:

##### **NOR/PSRAM control:**

|                 |                    |
|-----------------|--------------------|
| Memory type     | LCD Interface      |
| Bank            | Bank 1 NOR/PSRAM 1 |
| Write operation | Enabled            |
| Write FIFO      | Enabled            |
| Extended mode   | Disabled           |

##### **NOR/PSRAM timing:**

|                                           |     |
|-------------------------------------------|-----|
| Address setup time in HCLK clock cycles   | 15  |
| Data setup time in HCLK clock cycles      | 255 |
| Bus turn around time in HCLK clock cycles | 15  |

### 5.2. RCC

#### **High Speed Clock (HSE): Crystal/Ceramic Resonator**

#### 5.2.1. Parameter Settings:

##### **System Parameters:**

|                   |                    |
|-------------------|--------------------|
| VDD voltage (V)   | 3.3                |
| Instruction Cache | Enabled            |
| Prefetch Buffer   | Enabled            |
| Data Cache        | Enabled            |
| Flash Latency(WS) | 5 WS (6 CPU cycle) |

##### **RCC Parameters:**

|                         |          |
|-------------------------|----------|
| HSI Calibration Value   | 16       |
| TIM Prescaler Selection | Disabled |

|                                |                                 |
|--------------------------------|---------------------------------|
| HSE Startup Timeout Value (ms) | 100                             |
| LSE Startup Timeout Value (ms) | 5000                            |
| <b>Power Parameters:</b>       |                                 |
| Power Regulator Voltage Scale  | Power Regulator Voltage Scale 1 |
| Power Over Drive               | Enabled                         |

## 5.3. USART1

Mode: Asynchronous

### 5.3.1. Parameter Settings:

|                             |                           |
|-----------------------------|---------------------------|
| <b>Basic Parameters:</b>    |                           |
| Baud Rate                   | 115200                    |
| Word Length                 | 8 Bits (including Parity) |
| Parity                      | None                      |
| Stop Bits                   | 1                         |
| <b>Advanced Parameters:</b> |                           |
| Data Direction              | Receive and Transmit      |
| Over Sampling               | 16 Samples                |

\* User modified value

## 6. System Configuration

### 6.1. GPIO configuration

| IP     | Pin         | Signal      | GPIO mode                    | GPIO pull/up pull down      | Max Speed      | User Label   |
|--------|-------------|-------------|------------------------------|-----------------------------|----------------|--------------|
| FMC    | PE7         | FMC_D4      | Alternate Function Push Pull | No pull-up and no pull-down | Very High      |              |
|        | PE8         | FMC_D5      | Alternate Function Push Pull | No pull-up and no pull-down | Very High      |              |
|        | PE9         | FMC_D6      | Alternate Function Push Pull | No pull-up and no pull-down | Very High      |              |
|        | PE10        | FMC_D7      | Alternate Function Push Pull | No pull-up and no pull-down | Very High      |              |
|        | PE11        | FMC_D8      | Alternate Function Push Pull | No pull-up and no pull-down | Very High      |              |
|        | PE12        | FMC_D9      | Alternate Function Push Pull | No pull-up and no pull-down | Very High      |              |
|        | PE13        | FMC_D10     | Alternate Function Push Pull | No pull-up and no pull-down | Very High      |              |
|        | PE14        | FMC_D11     | Alternate Function Push Pull | No pull-up and no pull-down | Very High      |              |
|        | PE15        | FMC_D12     | Alternate Function Push Pull | No pull-up and no pull-down | Very High      |              |
|        | PD8         | FMC_D13     | Alternate Function Push Pull | No pull-up and no pull-down | Very High      |              |
|        | PD9         | FMC_D14     | Alternate Function Push Pull | No pull-up and no pull-down | Very High      |              |
|        | PD10        | FMC_D15     | Alternate Function Push Pull | No pull-up and no pull-down | Very High      |              |
|        | PD11        | FMC_A16     | Alternate Function Push Pull | No pull-up and no pull-down | Very High      |              |
|        | PD14        | FMC_D0      | Alternate Function Push Pull | No pull-up and no pull-down | Very High      |              |
|        | PD15        | FMC_D1      | Alternate Function Push Pull | No pull-up and no pull-down | Very High      |              |
|        | PD0         | FMC_D2      | Alternate Function Push Pull | No pull-up and no pull-down | Very High      |              |
|        | PD1         | FMC_D3      | Alternate Function Push Pull | No pull-up and no pull-down | Very High      |              |
|        | PD4         | FMC_NOE     | Alternate Function Push Pull | No pull-up and no pull-down | Very High      |              |
|        | PD5         | FMC_NWE     | Alternate Function Push Pull | No pull-up and no pull-down | Very High      |              |
|        | PD7         | FMC_NE1     | Alternate Function Push Pull | No pull-up and no pull-down | Very High      |              |
| RCC    | PH0-OSC_IN  | RCC_OSC_IN  | n/a                          | n/a                         | n/a            |              |
|        | PH1-OSC_OUT | RCC_OSC_OUT | n/a                          | n/a                         | n/a            |              |
| USART1 | PA9         | USART1_TX   | Alternate Function Push Pull | Pull-up                     | Very High<br>* |              |
|        | PA10        | USART1_RX   | Alternate Function Push Pull | Pull-up                     | Very High<br>* |              |
| GPIO   | PE3         | GPIO_Input  | Input mode                   | No pull-up and no pull-down | n/a            | BOOT_TRIGGER |
|        | PD2         | GPIO_Output | Output Push Pull             | No pull-up and no pull-down | Low            | EBI_RESET    |
|        | PD3         | GPIO_Output | Output Push Pull             | No pull-up and no pull-down | Low            | BL_EN        |

## ***6.2. DMA configuration***

nothing configured in DMA service

### 6.3. NVIC configuration

| Interrupt Table                         | Enable | Preenmption Priority | SubPriority |
|-----------------------------------------|--------|----------------------|-------------|
| Non maskable interrupt                  | true   | 0                    | 0           |
| Hard fault interrupt                    | true   | 0                    | 0           |
| Memory management fault                 | true   | 0                    | 0           |
| Pre-fetch fault, memory access fault    | true   | 0                    | 0           |
| Undefined instruction or illegal state  | true   | 0                    | 0           |
| System service call via SWI instruction | true   | 0                    | 0           |
| Debug monitor                           | true   | 0                    | 0           |
| Pendable request for system service     | true   | 0                    | 0           |
| System tick timer                       | true   | 0                    | 0           |
| USART1 global interrupt                 | true   | 0                    | 0           |
| PVD interrupt through EXTI line 16      | unused |                      |             |
| Flash global interrupt                  | unused |                      |             |
| RCC global interrupt                    | unused |                      |             |
| FPU global interrupt                    | unused |                      |             |

\* User modified value

## 7. Power Consumption Calculator report

### 7.1. Microcontroller Selection

|           |               |
|-----------|---------------|
| Series    | STM32F4       |
| Line      | STM32F446     |
| MCU       | STM32F446VETx |
| Datasheet | 027107 Rev5   |

### 7.2. Parameter Selection

|             |      |
|-------------|------|
| Temperature | 25   |
| Vdd         | null |

## 8. Software Project

### 8.1. Project Settings

| Name                              | Value                          |
|-----------------------------------|--------------------------------|
| Project Name                      | bootloader_st                  |
| Project Folder                    | E:\stm32f446vet6\bootloader_st |
| Toolchain / IDE                   | EWARM                          |
| Firmware Package Name and Version | STM32Cube FW_F4 V1.16.0        |

### 8.2. Code Generation Settings

| Name                                                            | Value                                 |
|-----------------------------------------------------------------|---------------------------------------|
| STM32Cube Firmware Library Package                              | Copy only the necessary library files |
| Generate peripheral initialization as a pair of '.c/.h' files   | No                                    |
| Backup previously generated files when re-generating            | No                                    |
| Delete previously generated files when not re-generated         | Yes                                   |
| Set all free pins as analog (to optimize the power consumption) | No                                    |